My Experience as a Teaching Assistant

Responsibilities, everyday decisions, and opportunities to grow

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Outline

Introduction	My experience supporting physics courses
TA Responsibilities	What a teaching assistant need to do
My Experience in Practice	How I approached the work
Reflections and Growth	What I gained as a TA

My TA Experience

Physics courses I supported

General Physics I — Fall 2023

Advanced Quantum Mechanics — Fall 2024

Quantum Mechanics — Fall 2025

My main responsibilities

Grading homework

Preparing tutorials

Reviewing and grading exams

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TA Work Across a Semester

The exact responsibilities depend on the course and the instructor.

Before the semester

Clarify duties with the instructor.

Set up a class communication channel.

Ensure every student can receive announcements.

During the semester

Grade homework; prepare solutions if needed.

Answer students' questions.

Prepare tutorials on the agreed schedule.

Around exams

Review the exam paper in advance.

Define grading criteria.

Grade the exam scripts.

Stay in touch with the instructor throughout the semester.

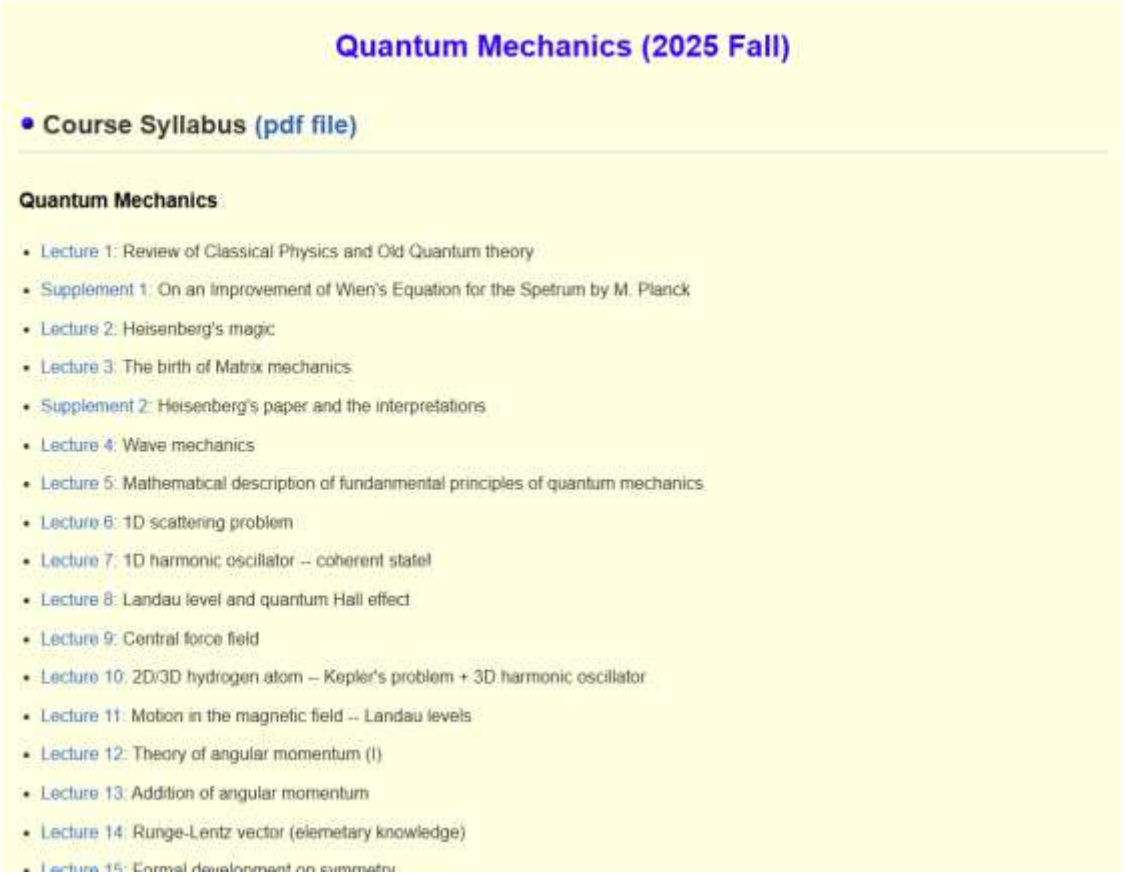
Preparing for the Semester

Agree on expectations early

I contacted the instructor before the semester to understand what was needed.

Created a course website for lecture notes and homework assignments.

Attended the first class to help with course setup.



Quantum Mechanics (2025 Fall)

- Course Syllabus (pdf file)

Quantum Mechanics

- Lecture 1: Review of Classical Physics and Old Quantum theory
- Supplement 1: On an Improvement of Wien's Equation for the Spetrum by M. Planck
- Lecture 2: Heisenberg's magic
- Lecture 3: The birth of Matrix mechanics
- Supplement 2: Heisenberg's paper and the interpretations
- Lecture 4: Wave mechanics
- Lecture 5: Mathematical description of fundanmental principles of quantum mechanics
- Lecture 6: 1D scattering problem
- Lecture 7: 1D harmonic oscillator -- coherent state
- Lecture 8: Landau level and quantum Hall effect
- Lecture 9: Central force field
- Lecture 10: 2D/3D hydrogen atom -- Kepler's problem + 3D harmonic oscillator
- Lecture 11: Motion in the magnetic field -- Landau levels
- Lecture 12: Theory of angular momentum (I)
- Lecture 13: Addition of angular momentum
- Lecture 14: Runge-Lentz vector (elemetary knowledge)
- Lecture 15: Formal development on symmetry

Starting the First Class

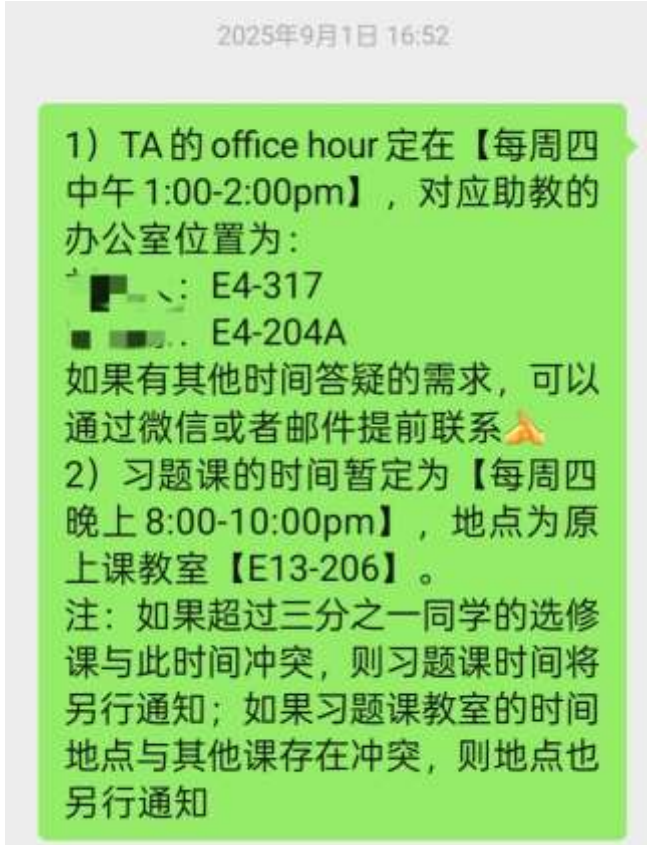
Make communication easy

Set up a WeChat group during the first class.

Consider iWestlake or email as alternative channels.

Share key information: office hours, tutorials, and homework arrangements.





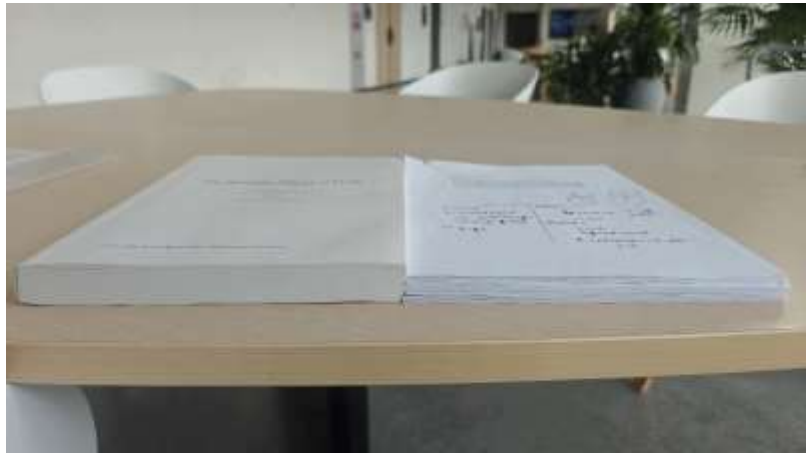
Homework Grading and Tutorials

Read the reasoning, not just the answer

Work out the solutions before grading.

Set clear grading criteria.





Homework Grading and Tutorials

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Work out the solutions before grading.

Set clear grading criteria; Check reasoning

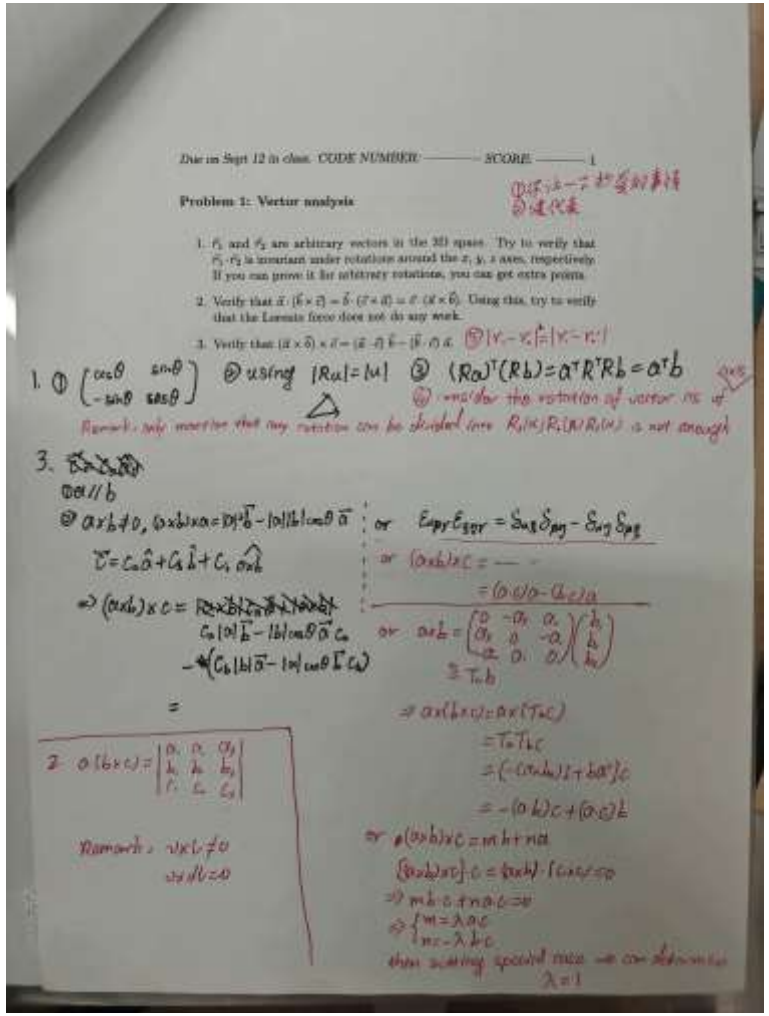
Add brief comments explaining errors.

Accept valid alternative solutions.

Discuss common mistakes in tutorials.

Share interesting solutions anonymously.

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Preparing and Grading Exams

Review the paper before the exam

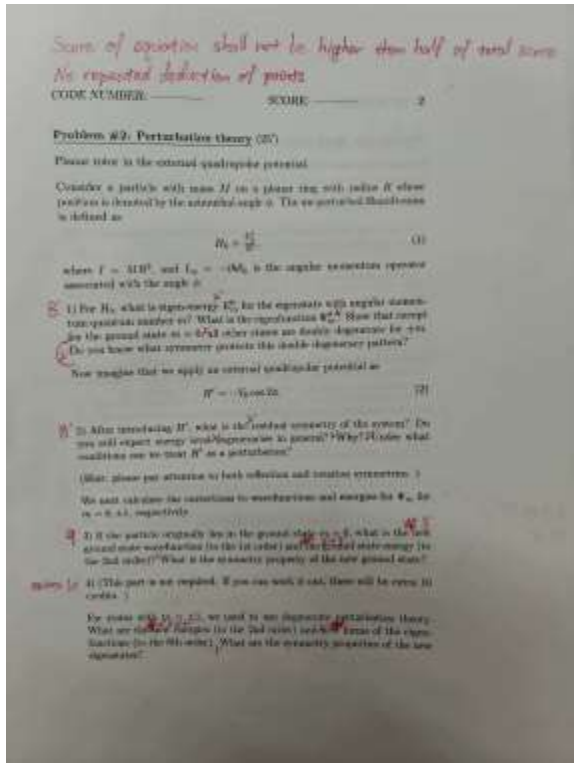
Solve the paper; check for errors and ambiguity.

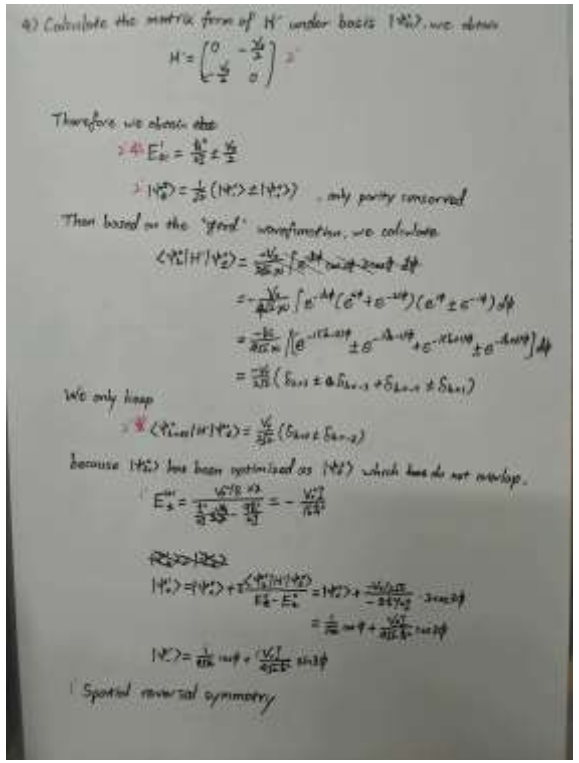
Discuss the difficulty with the instructor.

Set grading criteria for the revised paper.

Coordinate exam logistics with the teaching administrator.

Invigilate, and grade consistently.





What I Gained as a TA

Connections between teaching and research

Practice in explaining ideas and listening

Opportunities to revisit core concepts

Finding Meaning in the Role

Student appreciation

Students’ appreciation made the effort worthwhile.

Ordinary, careful work can be meaningful.





THANK YOU

QUESTIONS & DISCUSSION

西湖大學
WESTLAKE UNIVERSITY